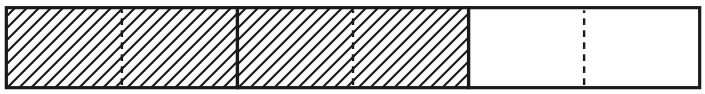
Dividing Fractions

How many of these fit in that? Make the pieces the same size.

2/3 ÷ 1/6: how many sixths fit in two thirds?



Cut the thirds into sixths.
The shaded part is 4 sixths.

$$\frac{2}{3} \div \frac{1}{6} = 4$$

Same bottoms: divide the tops

$$\frac{3}{4} \div \frac{1}{8} = \frac{6}{8} \div \frac{1}{8} = 6$$

When the pieces are the same size, it is 6 pieces ÷ 1 piece.

It doesn't have to come out even

$$\frac{3}{4} \div \frac{2}{4} = \frac{3}{2} = 1\frac{1}{2}$$

3 pieces ÷ 2 pieces:
one and a half twos fit.

Your turn!

$$\frac{1}{2} \div \frac{1}{8} = \boxed{4}$$

For teachers

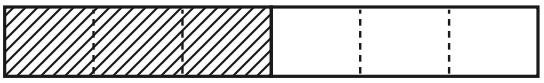
Goal: divide fractions by fractions (6.NS.1). It builds on packet 84 (how many pieces fit). Say “Make the pieces the same size, then divide the tops.”

Dividing Fractions: Practice

Same size pieces, then divide the tops.

Part 1 How many fit? The strip is cut into the smaller pieces.

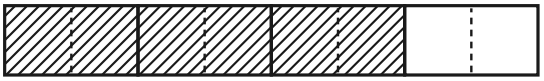
1)



$\frac{1}{2} \div \frac{1}{6} = \boxed{3}$

How many pieces of 1/6 fit in the shaded part?

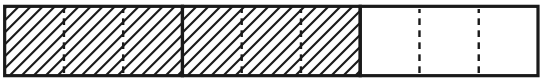
2)



$\frac{3}{4} \div \frac{1}{8} = \boxed{6}$

How many pieces of 1/8 fit in the shaded part?

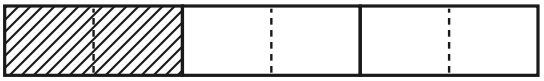
3)



$\frac{2}{3} \div \frac{1}{9} = \boxed{6}$

How many pieces of 1/9 fit in the shaded part?

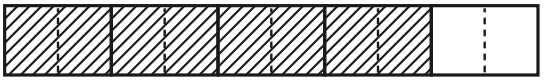
4)



$\frac{1}{3} \div \frac{1}{6} = \boxed{2}$

How many pieces of 1/6 fit in the shaded part?

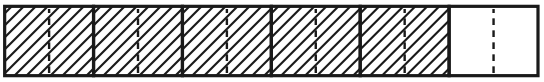
5)



$\frac{4}{5} \div \frac{1}{10} = \boxed{8}$

How many pieces of 1/10 fit in the shaded part?

6)



$\frac{5}{6} \div \frac{1}{12} = \boxed{10}$

How many pieces of 1/12 fit in the shaded part?

Dividing Fractions: Practice (continued)

Part 2 Same bottoms first Write both with the same bottom, then divide the tops.

7) $\frac{1}{2} \div \frac{1}{8} = \frac{\boxed{4}}{8} \div \frac{1}{8} = \boxed{4}$

8) $\frac{2}{3} \div \frac{2}{9} = \frac{\boxed{6}}{9} \div \frac{2}{9} = \boxed{3}$

9) $\frac{3}{5} \div \frac{3}{10} = \frac{\boxed{6}}{10} \div \frac{3}{10} = \boxed{2}$

10) $\frac{5}{6} \div \frac{1}{12} = \frac{\boxed{10}}{12} \div \frac{1}{12} = \boxed{10}$

11) $\frac{3}{4} \div \frac{3}{8} = \frac{\boxed{6}}{8} \div \frac{3}{8} = \boxed{2}$

12) $\frac{1}{4} \div \frac{1}{12} = \frac{\boxed{3}}{12} \div \frac{1}{12} = \boxed{3}$

13) $\frac{2}{5} \div \frac{1}{10} = \frac{\boxed{4}}{10} \div \frac{1}{10} = \boxed{4}$

14) $\frac{7}{8} \div \frac{1}{16} = \frac{\boxed{14}}{16} \div \frac{1}{16} = \boxed{14}$

Part 3 Divide No help this time.

15) $\frac{4}{5} \div \frac{1}{5} = \boxed{4}$

16) $\frac{6}{7} \div \frac{2}{7} = \boxed{3}$

17) $\frac{5}{8} \div \frac{1}{8} = \boxed{5}$

Dividing Fractions: Practice (continued)

Part 3 Divide (continued)

18) $\frac{9}{10} \div \frac{3}{10} = \boxed{3}$

19) $\frac{2}{3} \div \frac{1}{12} = \boxed{8}$

20) $\frac{3}{4} \div \frac{1}{16} = \boxed{12}$

21) $\frac{1}{2} \div \frac{1}{10} = \boxed{5}$

22) $\frac{5}{6} \div \frac{5}{12} = \boxed{2}$

23) $\frac{4}{9} \div \frac{2}{9} = \boxed{2}$

24) $\frac{7}{10} \div \frac{7}{20} = \boxed{2}$

25) $\frac{2}{5} \div \frac{2}{15} = \boxed{3}$

26) $\frac{3}{8} \div \frac{3}{16} = \boxed{2}$

Part 4 When it isn't even No help this time.

27) $\frac{3}{4} \div \frac{2}{4} = \frac{\boxed{3}}{\boxed{2}}$

28) $\frac{5}{6} \div \frac{2}{6} = \frac{\boxed{5}}{\boxed{2}}$

Dividing Fractions: Practice (continued)

Part 4 When it isn't even (continued)

29) $\frac{2}{3} \div \frac{3}{4} = \frac{8}{9}$

30) $\frac{4}{5} \div \frac{2}{3} = \frac{6}{5}$

31) $\frac{1}{2} \div \frac{3}{4} = \frac{2}{3}$

32) $\frac{7}{8} \div \frac{1}{4} = \frac{7}{2}$

33) $\frac{5}{9} \div \frac{2}{3} = \frac{5}{6}$

34) $\frac{3}{10} \div \frac{1}{4} = \frac{6}{5}$

35) $\frac{5}{6} \div \frac{3}{8} = \frac{20}{9}$

36) $\frac{1}{3} \div \frac{2}{5} = \frac{5}{6}$

37) $\frac{9}{10} \div \frac{3}{4} = \frac{6}{5}$

38) $\frac{2}{7} \div \frac{1}{2} = \frac{4}{7}$

Dividing Fractions: Practice (continued)

Part 5 Whole numbers and fractions No help this time.

39) $3 \div \frac{3}{4} = 4$

40) $4 \div \frac{2}{5} = 10$

41) $6 \div \frac{3}{4} = 8$

42) $2 \div \frac{2}{3} = 3$

43) $5 \div \frac{5}{6} = 6$

44) $8 \div \frac{2}{3} = 12$

45) $3 \div \frac{3}{8} = 8$

46) $6 \div \frac{2}{7} = 21$

Part 6 More dividing

47) $\frac{2}{3} \div \frac{2}{3} = 1$

48) $\frac{7}{8} \div \frac{4}{8} = \frac{7}{4}$

49) $\frac{2}{5} \div \frac{1}{6} = \frac{12}{5}$

Dividing Fractions: Practice (continued)

Part 6 More dividing (continued)

50) $\frac{5}{6} \div \frac{1}{2} = \frac{5}{3}$

51) $\frac{2}{3} \div \frac{7}{10} = \frac{20}{21}$

52) $\frac{1}{2} \div \frac{4}{10} = \frac{5}{4}$

53) $\frac{3}{6} \div \frac{1}{3} = \frac{3}{2}$

54) $\frac{1}{3} \div \frac{7}{12} = \frac{4}{7}$

55) $\frac{2}{4} \div \frac{1}{3} = \frac{3}{2}$

56) $\frac{1}{3} \div \frac{1}{6} = 2$

57) $\frac{6}{8} \div \frac{2}{12} = \frac{9}{2}$

58) $\frac{2}{4} \div \frac{3}{5} = \frac{5}{6}$

59) $\frac{7}{8} \div \frac{2}{3} = \frac{21}{16}$

60) $\frac{2}{4} \div \frac{4}{10} = \frac{5}{4}$

61) $\frac{3}{8} \div \frac{6}{8} = \frac{1}{2}$

Dividing Fractions: Practice (continued)

Part 6 More dividing (continued)

62) $\frac{8}{10} \div \frac{5}{8} = \frac{32}{25}$

63) $\frac{4}{5} \div \frac{6}{8} = \frac{16}{15}$

64) $\frac{1}{2} \div \frac{3}{5} = \frac{5}{6}$

65) $\frac{7}{8} \div \frac{10}{12} = \frac{21}{20}$

66) $\frac{1}{2} \div \frac{2}{3} = \frac{3}{4}$

67) $\frac{1}{2} \div \frac{4}{6} = \frac{3}{4}$

68) $\frac{1}{3} \div \frac{3}{4} = \frac{4}{9}$

69) $\frac{1}{2} \div \frac{2}{8} = 2$

70) $\frac{1}{3} \div \frac{3}{8} = \frac{8}{9}$

Part 7 More whole numbers

71) $4 \div \frac{4}{5} = 5$

72) $5 \div \frac{3}{9} = 15$

73) $4 \div \frac{2}{3} = 6$

Dividing Fractions: Practice (continued)

Part 7 More whole numbers (continued)

74) $9 \div \frac{5}{10} = 18$

75) $4 \div \frac{5}{10} = 8$

76) $6 \div \frac{1}{8} = 48$

77) $4 \div \frac{3}{9} = 12$

78) $3 \div \frac{3}{5} = 5$

79) $4 \div \frac{1}{3} = 12$

80) $7 \div \frac{3}{6} = 14$

81) $2 \div \frac{2}{10} = 10$

82) $4 \div \frac{2}{4} = 8$

Part 8 Remember? Problems from earlier packets.

83) 25% of 84 = 21

84) 6 notebooks for $9.00 = $1.50 each

Dividing Fractions: Practice (continued)

Part 8 Remember? (continued)

85) $6 \times \frac{2}{9} = \frac{12}{9}$

86) $4.2 \times 3 = 12.6$

87) $12 : 5 = 36 : 15$

88) $6.04 < 6.4$

Part 9 Word problems How many fit?

89) A ribbon is 3/4 m long. Each bow uses 1/8 m. How many bows can you make?

Number sentence 3/4 ÷ 1/8 = 6

Answer 6 bows

Draw or show your work

90) You have 2/3 cup of flour. A batch uses 1/6 cup. How many batches can you make?

Number sentence 2/3 ÷ 1/6 = 4

Answer 4 batches

Draw or show your work

91) A 6 m rope is cut into pieces 3/4 m long. How many pieces?

Number sentence 6 ÷ 3/4 = 8

Answer 8 pieces

Draw or show your work

Dividing Fractions: Practice (continued)

Part 9 Word problems (continued)

92) A jug holds 5/6 L. A glass holds 1/3 L. How many glasses does the jug fill?

Number sentence **5/6 ÷ 1/3 = 2 1/2**

Answer **2 1/2** glasses

Draw or show your work

93) Challenge: A cake needs 2/3 cup sugar. A scoop is 1/6 cup. You added 2 scoops. How many more?

Number sentence **2/3 ÷ 1/6 = 4, 4 − 2 = 2**

Answer **2** scoops

Draw or show your work

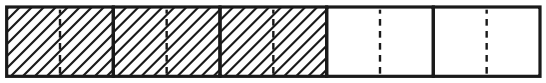
Check-In: Dividing Fractions

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 How many fit?

1)



$\frac{3}{5} \div \frac{1}{10} = 6$

How many pieces of 1/10 fit in the shaded part?

Part 2 Divide

2) $\frac{5}{6} \div \frac{2}{3} = \frac{5}{4}$

3) $\frac{3}{5} \div \frac{1}{10} = 6$

4) $\frac{7}{9} \div \frac{7}{18} = 2$

5) $\frac{4}{5} \div \frac{3}{10} = \frac{8}{3}$

Part 3 Whole numbers and fractions

6) $10 \div \frac{5}{6} = 12$

7) $6 \div \frac{2}{3} = 9$